



CI4305 COURSEWORK 1

App Design Prototype Project (Part 1)

Deadline: 13/12/2024

Group Members and Participation

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CONTENTS

1.1 – 1.3 Introduction and background research	3
1.4 – 1.5 Stakeholders.....	6
1.6 – 1.7 Requirements elicitation	7
1.8 – 1.9 Requirements AND NON-FUNCTIONAL REQUIREMENTS	9
1.10 – 1.12 Requirements MODELLING, VALIDATION & DOCUMENTATION	0
1.13 User PERSONA	5
1.14 User Stories	7
1.15 – 1.16 Navigation Design	8
1.17 – 1.20 Wireframing (Maximum of 5)	10

1.1 – 1.3 INTRODUCTION AND BACKGROUND RESEARCH

Complete this section in accordance with the coursework marking scheme.

Page Specification: Maximum 3 Pages.

Application Name

Gamers Portal

The name we gave for our application is meant to be related to the content that we are going to have. Since we are going to advertise games and game-related content, we decided that Gamers Portal would be the most suitable and catchy for this occasion.

Application Prototype

Games Online Retail Store

During our initial meeting, we collectively agreed to pursue a project within our comfort zone, something we're familiar with. After discussing our experiences, we decided to create a Games Online Retail Store, taking advantage of our knowledge of gaming platforms and considering what we would like to improve in existing online retail stores.

Target Devices

Smartphone, Tablet, Desktop and Laptop

Developing our application for multiple devices ensures that our application is accessible to a broader audience. The application would be used to access games, game-related merchandise, streaming content and events. Users can download and install the app on their chosen device to access free games, purchase games and related merchandise, view content creator streams, view and stream special events, download and launch games acquired. Additionally, we plan to offer a web browser version of the application, which will allow users to do everything described above apart from downloading and launching games.

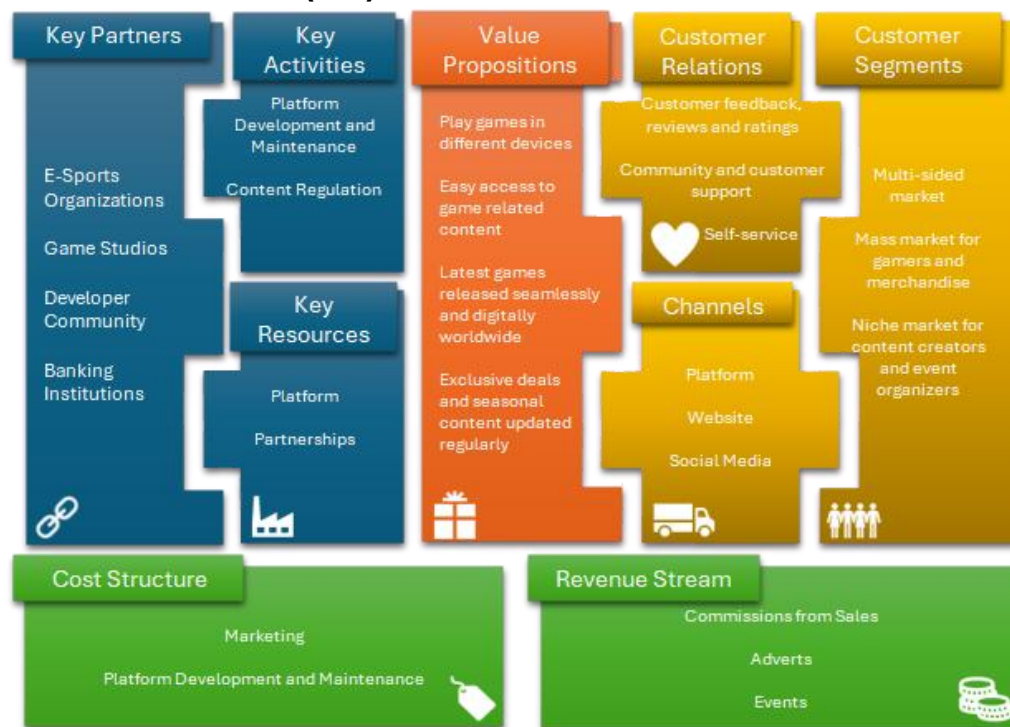
Revenue Model

Freemium and Commissions

Our application will generate revenue through three primary channels.

1. **Freemium:** This approach allows us to offer the application to a wider audience free of charge. When freemium users try to access content, for example when they access streams an advert will appear. This way we can generate more revenue exposing more people to advertising.
2. **Commissions:** We will partner with game and merchandise sellers, allowing them to promote their products through our platform. We will then take a commission on all products sold through the platform.

Business Model Canvas (1.2)



Business Research

Amazon, Steam and Apple Store

This business research explores three major platforms, Amazon, Steam and Apple Store. It points out their strengths and weaknesses and help us evaluate where our model can do better.

Amazon's primary source of income is from retail sales and a freemium subscription revenue model that uses Amazon Prime, which offers benefits like free shipping, exclusive deals and access to the Amazon video streaming platform.

Benefits:

Reliability: Amazon's user base reviews system creates a level of trust and reliability. Verified sellers brings confidence to users that wish to purchase products through the Amazon platform.

Convenience: Amazon extensive online products list and delivery service makes shopping highly accessible and efficient, bringing products directly to customers front door.

Limitations:

Ethical Concerns: Amazon has faced criticism over environmental impact and working conditions, potentially deterring customers from purchasing products due to ethical concerns.

Privacy: Amazon collects huge amounts of user data for advertising, raising concerns about user privacy and data security.

Steam is a leading digital distribution platform for video games, developed by Valve Corporation. It offers a large user base and a comprehensive ecosystem for both gamers and developers, providing numerous business advantages and some challenges.

Benefits:

Sales and Discounts: Steam frequently announces sales and discounts, which attracts more customers and boost game sales.

Developer Support: Steam provides a wide range of tools to support developers. Steam works, for example, offers features like achievements, cloud saves, and multiplayer support.

Limitations:

Revenue Share: Steam takes a significant cut of the revenue from game sales, typically around 30%, making it a challenge especially for indie developers.

Dependency on Platform: Relying heavily on Steam can be risky for developers if the platform changes its policies or if there are technical issues.

Apple Store

Apple Store is widely used and available for its users across many different regions, connecting over 650 million people each week. The Apple Store is free to use but it offers a range of paid services such as Apple Arcade, Apple Fitness+, Apple Music between many others.

Benefits:

Global Accessibility: Apple manages international taxes and currency, so it handles any international tax complications, allowing users easy access to global markets.

Limitations:

Costly: Apple commission fee is set at 30%, which makes it less lucrative for users compared to Microsoft Store 15% fee or Google Play Store fee between 15% and 30%.

Limited Flexibility: Pricing and Apple's policies make it restrictive to users.

Conclusion and Competitive Advantage:

Our model aims to have a competitive edge by addressing key areas where Amazon, Steam and Apple Store have limitations. We intend to focus on developer support initiatives, better ethical practices and an enhanced user experience.

Developer Support: By adopting a more equitable revenue-sharing model and providing better support for indie developers, we can attract a more diverse content creator user base. Our model will allow smaller developers to retain more of their earnings, differentiating us from Steam and Apple Store.

Ethical Practices: By prioritizing a more transparent and user-friendly data policies and by adopting better sustainability practises, we address the ethical concerns that challenges Amazon reputation.

User Experience: Improving algorithms and adding merchandise and events content, will enhance user experience by giving the user a broader offer of content available. This will give us an edge over Steam where users usually experience content stagnation, seeing similar content repeatedly.

1.4 – 1.5 STAKEHOLDERS

Complete this section in accordance with the coursework marking scheme.

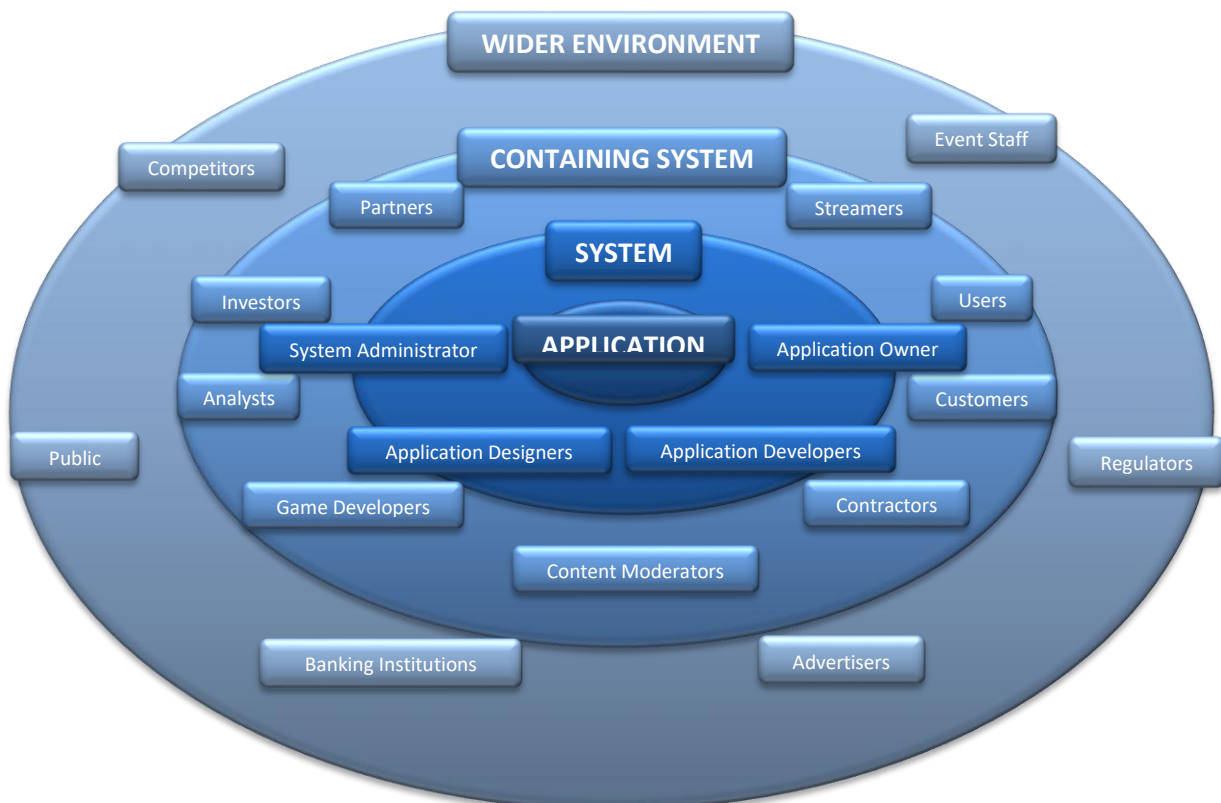
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Primary Stakeholder Analysis

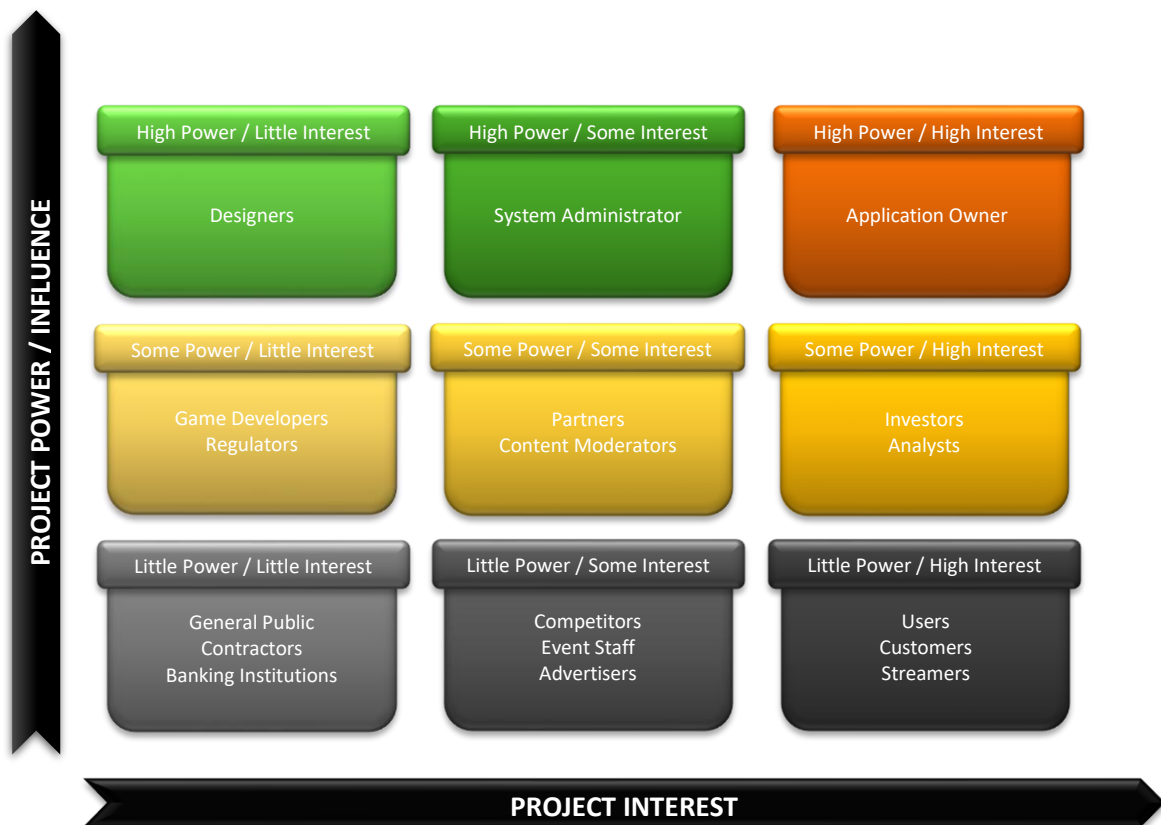
Below are a short list of primary stakeholders and reasons why they have been assigned their positions in the onion diagram.

1. Users: Engage in the application providing valuable feedback and generate revenue through advertising.
2. Customers: Main source of revenue they are the ones to pay for games, merchandise and event tickets.
3. Application Developers: Responsible for implementing and designing new functionalities and features, they have full access to the application back-end components.
4. Game developers: Use our service to market their own games and sell them. They generate the main content we sell on our platform.
5. System Administrator: Help maintain the application, making sure that user experience is the best and remains consistent and intuitive.
6. Investors: With their interest on the application performance and success they fund the application development and running cost.
7. Competitors: Drive continuous innovation and platform improvement by consistently challenging the application to maintain its position as the best in the market.

Stakeholder Onion Diagram



Stakeholder Power Interest Grid



1.6 – 1.7 REQUIREMENTS ELICITATION

Complete this section in accordance with the coursework marking scheme.

Page Specification: Maximum 2 Pages.

Elicitation Technique

To understand stakeholders' expectations for the application, this elicitation technique will focus on in-depth, open-ended questioning. This approach allows stakeholders to express themselves freely and enable us to explore potential improvements on features from those who interact with the application regularly.

Methods:

Customer Survey – We will conduct a survey with open-ended questions, allowing users to freely express their opinions. The survey will be structured to gain insights into users' experiences, identify areas that users feel improvement is required within the application, understand how users value the application and what additional features and services they'd like to be included in the application, and collect personal suggestions on the application's future developments.

Staff Interviews – In one-on-one interviews, we will engage with staff members using a flexible, semi-structured format. This will allow the use of a base set of questions as a guide to ensure we cover essential topics, adjust questions on the go to dig deeper into specific points, and gain a better understanding of internal processes, challenges, and opportunities for improvement from the staff's perspective.

Elicitation Questions

For Customer Surveys

1. What are the primary activities you perform in the application?
2. What are the most valuable aspects of the application for you?
3. How do you feel about the features and services that have been provided?
4. What changes would make the application more useful to you?
5. How do you envision the application in the future to better meet your needs?

For Staff Interviews

1. In your role, what areas of the application do you interact the most, and what challenges do you face?
2. Are there specific tools or features that would make your job easier or more efficient?
3. What additional value could we provide to users that isn't currently being addressed?
4. What is your long-term vision for the platform, and what suggestions do you have for its future development?
5. Can you think of any possible risks or challenges that could arise, and if so, mitigation strategies that we could implement?

1.8 – 1.9 REQUIREMENTS AND NON-FUNCTIONAL REQUIREMENTS

Complete this section in accordance with the coursework marking scheme.

Page Specification: Maximum 1 Page.

Functional Requirements

Examples of User Functional Requirements:

User Management:

Log in

Create Account

Manage Account

Delete Account

Create Profile

Products Library:

View Events

Purchase Games

View Suggestions

Purchase Merchandise

Launch Games

Content Management:

View Reviews

Post Reviews

Update Content

Administrator Functional Requirements:

User Management:

Deactivate User

Delete User

Reset Password

System Moderation:

Confirm User

Log Activity

Delete Content

Ban Content

System Management:

Log Errors

Generate Reports

Manage Database

Non-Functional Requirements

Log in – System must process the login request within 2 seconds 95% of the time to ensure a reliable user experience.

Confirm user – Email must be sent after user account creation within 5 to 10 seconds to confirm user email to activate account.

View suggestions – Application suggestions list must change every 24 hours giving the user more options of content for purchase.

View events – Application events on display must be available 99% of the time ensuring user have accurate details at any time.

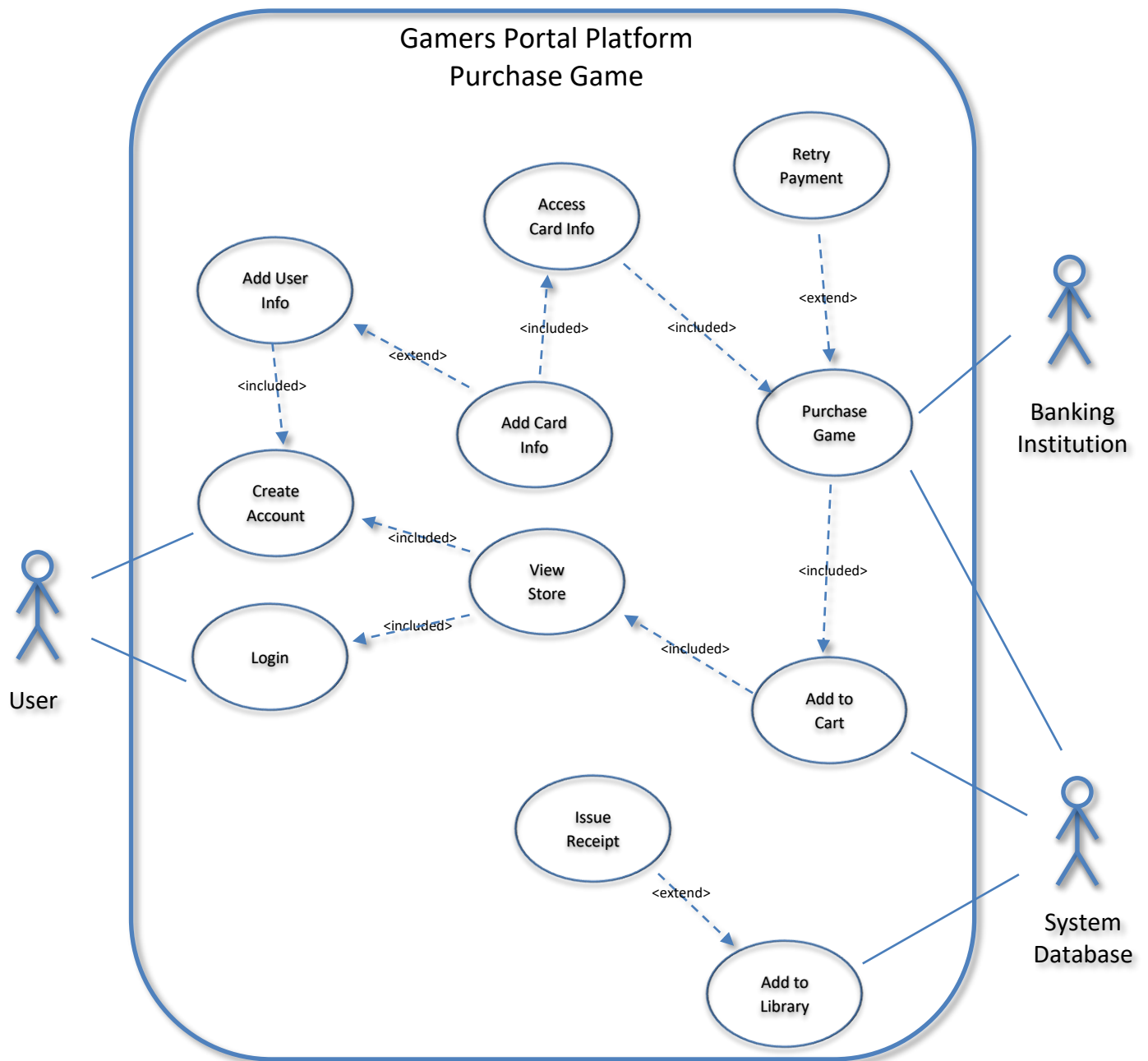
Purchase games – System must send a confirmation email within 30 seconds after the purchase, even under peak moment time.

1.10 – 1.12 REQUIREMENTS MODELLING, VALIDATION & DOCUMENTATION

Complete this section in accordance with the coursework marking scheme.

Page Specification: Maximum 5 Pages.

Use Case Diagram



Use Case Description

Use Case Description		
Use Case: 1.1	Process Purchase Game	
Goal in Context	Platform users are able own and pay for games	
Scope & Level	Products Library, User	
Preconditions	User logged in, system online	
Success End Condition	Payment processed, receipt issued, game added to user library	
Failed End Condition	Payment not processed, cancelled transaction	
Primary, Secondary Actors	User (U) System Database (SD) Banking Institution (BI)	
Triggers	Purchase Game	
Description	Step	Action
	1	U logs in or creates account
	2	U searches a game on the store
	3	U adds game to cart
	4	U places a buy order on the platform
	5	SD takes payment information
	6	BI takes payment
	7	SD adds game to user products library
	8	SD issues a receipt for transaction
Extensions	Step	Branching Action
	4a	If payment information not stored, ask user for payment information
	6a	If payment not successful, notify user and ask to try again
Related Information		
Priority	Must	
Performance	<5s response from the system	
Frequency	Up to 15 per minute	
Channels to Actors	Online	
Open Issues		
Due Date		
...any other management information...		
Included In		
Inclusions		

Requirements Catalogues

Requirement Catalogue Entry			
Source: Catherine	Priority: High	Owner: System Administrator	Requirement ID: FR.02.v01

Functional requirement	Create Account
The user must be able to create and account through the application or website, providing a valid email address for validation and safe password for security.	

Benefits:
Enable users to have access to the products and services provided in the application and give users a more personalized type of experience.
Comments/ Suggested solutions:
Use proper encryption methods to ensure a safe user experience.
Make the process easy and without unnecessary steps so it does not discourage the user.
Related documents:
Use Case: Create Account
Classes: Accounts
Related requirement:
FR.01 Manage Account
Resolution: Implemented

Non-functional component			
Description	Target value	Acceptable range	Comments
Confirming User	5 to 10 seconds	30 seconds	Email must be sent to user to confirm the user email to activate account.
Creation Time	3 to 5 minutes	10 minutes	This need to be easy and concise so users can create an account within a short amount of time.

Requirement Catalogue Entry

Source: Michael

Priority: Must

Owner: System
Administrator

Requirement ID: FR.04v0.3

Functional requirement**Purchase Game**

The user must be able to purchase any content from the application or webpage, using a secure and easy check out process.

Benefits:

Allows users to access and purchase content from anywhere increasing revenue and encouraging the purchase of more content.

Comments/ Suggested solutions:

Use a secure payment gateway to manage payments securely and efficiently.
Design a simple checkout process to minimize drop-off rates.

Related documents:

Use case: Purchase Game
Classes: Payments and Products Library

Related requirement:

FR.03 Make Payments
FR.05 Manage Receipts

Resolution: Implemented**Non-functional component**

Description	Target value	Acceptable range	Comments
Email Verification	30 seconds	5 minutes	Need to send the user an email verification of the purchase.
Purchase time	5 to 10 seconds	20 seconds	Purchase should be quick for users with payment information already saved.

Requirement Catalogue Entry

Source: Chloe

Priority: High

Owner: Application
Designer

Requirement ID: FR.06v0.2

Functional requirement**Launch Games**

The user must be able to launch games they own through the application only. The application should check for device specification before launching the game and notify the user if their device does not meet the minimum requirements for a stable gameplay.

Benefits:

Gives user access to the content securely and prevents performance issue by alerting user of potential compatibility issues.

Comments/ Suggested solutions:

Only accessible through the application.

Related documents:

Use Case: Launch Game

Classes: Accounts and Products Library

Related requirement:

FR.04 Purchase Game

Resolution: Implemented

Non-functional component

Description	Target value	Acceptable range	Comments
Performance	30 seconds	1 minute	System must be able to open games effectively.
Reliability	3 seconds	6 seconds	If the system failed to open the game, it should produce a report about the error.

1.13 User PERSONA

Complete this section in accordance with the coursework marking scheme.

Page Specification: Complete the templates below.

Persona 1

NAME:

Jepeto Smith

BACKGROUND:

Jepeto is a 25-year-old single male gamer who is a HMV worker. He is not very sporty but has a keen interest in gaming and media. His hobbies include Gaming, Music, Art and Photography. Due to his job, he often interacts with lots of likeminded people who also enjoy different forms of media which makes him an expert.

TECHNOLOGY EXPERIENCE:

His internet usage is extremely broad and mainly related to his hobbies and interests. Jepeto has lots of friends and enjoys using the internet to play games online. He originally used to mainly play console games, but he is now mainly a PC Gamer. His new gaming habits require him to find a store in which he can easily and conveniently purchase and download games. He prefers a simple and straight forward approach that doesn't require much effort. Often after work Jepeto likes to play with friends online and usually chats and shares gaming info on community forums.

HABITS THAT RELATE TO YOUR APP:

Jepeto enjoys online shopping and has lots of experience using online stores such as Steam and Epic Games, but he is interested in looking for a platform with better services and functionalities such as social features and being able to buy merchandise. Jepeto also enjoys an interactive platform with features such as reviews and community discussions based on products being sold. He likes these features because it helps him properly decide what games and merch would be the most suitable for him. He often finds shopping in person overwhelming which leads to him preferring online shopping.

FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS:

Log in, Create Account, Manage Account, Delete Account, Create Profile, View Events, Purchase Games, View Suggestions, Purchase Merchandise, Launch Games, View Reviews, Post Reviews, Update Content.

Log in – System must process the login request within 2 seconds 95% of the time to ensure a reliable user experience.

Confirm user – Email must be sent after user account creation within 5 to 10 seconds to confirm user email to activate account.

View suggestions – Application suggestions list must change every 24 hours giving the user more options of content for purchase.

View events – Application events on display must be available 99% of the time ensuring user have accurate details at any time.

Purchase games – System must send a confirmation email within 30 seconds after the purchase, even under peak moment time

Persona 2:

NAME:

Rachel Williams

BACKGROUND:

Rachel is a 19-year-old female University student who is currently studying Computer Games Programming. She is highly enthusiastic about programming and designing different games as part of her personal projects outside school hours. Rachel enjoys sharing her ideas and discussing them with likeminded individuals online which has helped her to build connections through her interest in games programming.

TECHNOLOGY EXPERIENCE:

Rachel uses the internet for many things whether it be related to her interest in games programming or just general usage such as watching movies. However, she spends most of her time researching about existing games by reading reviews and playing them which allows her to study her competitors and help improve her ideas. Rachel wishes to find an app where she can easily do both so it can make her research more efficient and less complicated.

HABITS THAT RELATE TO YOUR APP:

Rachel does a lot of online shopping as she enjoys playing different games and to also study her competitors. An app that enables her to read many different reviews whilst being able to upload her games will help her to fulfil her intentions. Rachel prefers to shop both online as it is much more convenient and in person as it gives her a chance to gain an insight on the sellers'/shop owners' opinions about the games she is purchasing.

FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS:

Log in, Create Account, Manage Account, Delete Account, Create Profile, View Events, Purchase Games, View Suggestions, Purchase Merchandise, Launch Games, View Reviews, Post Reviews, Update Content.

Log in – System must process the login request within 2 seconds 95% of the time to ensure a reliable user experience.

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1.14 USER STORIES

Complete this section in accordance with the coursework marking scheme.

Page Specification: Maximum 1 Page.

User Stories

1. As an occasional gamer, I want the ability to purchase games through the application or the web app so I can have easy access to the game's library anywhere.
2. As a professional gamer, I want the ability to have recording tools in app so I can stream content in a more efficient way.
3. As a streamer, I want to be able to stream my content in the application and outside with ease, so I can reach a higher volume of audience.
4. As a game's developer, I want the ability to manage the games I uploaded in one place so this can make my job easier to manage.
5. As a content creator, I want to be able to have an online shop in the application so I can operate my content within the application like a shop.
6. As a content moderator, I want the ability to search reviews and text by keywords, so it becomes easier to find content that must be removed.
7. As an investor, I want to be able to see the application operations and prospects easier so I can better evaluate the business potential.
8. As a system administrator, I want the user to be able to create an account through the application or website be able to see all changes made on the system so I can better analyse what has been done in the application and what can be improved.

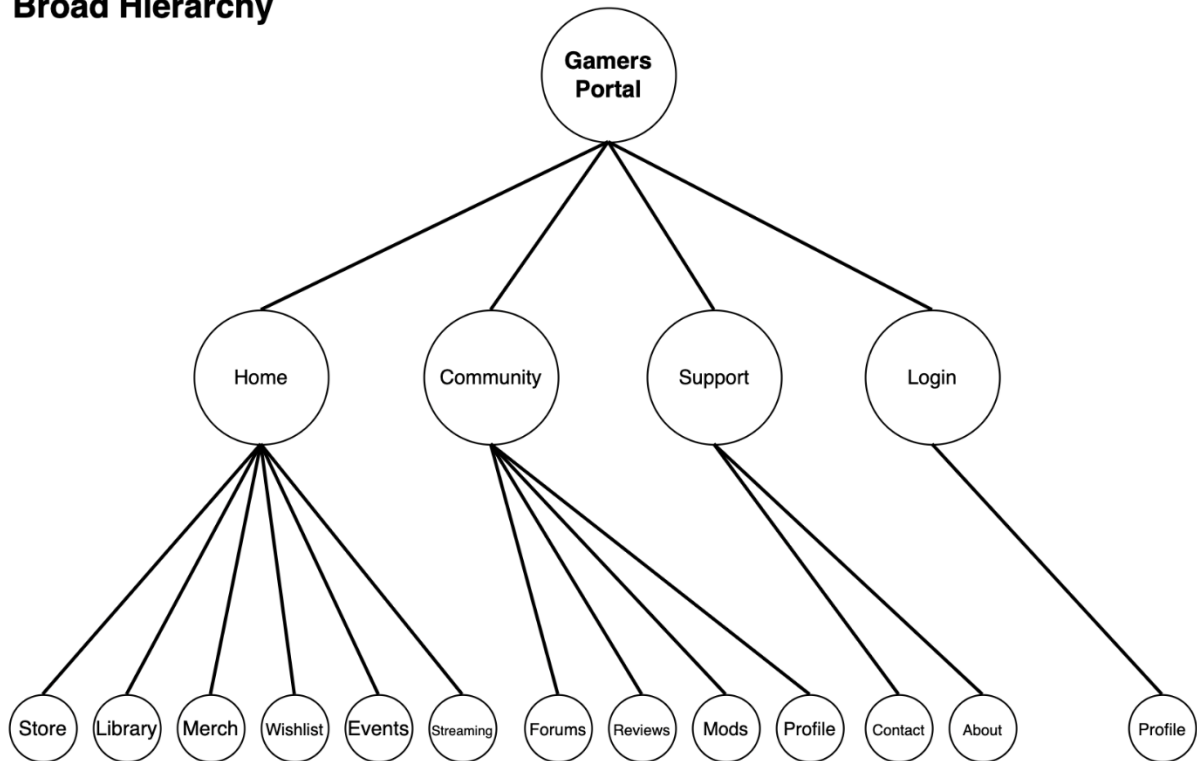
1.15 – 1.16 NAVIGATION DESIGN

Complete this section in accordance with the coursework marking scheme.

Page Specification: No Page Specification.

Navigation Design

Broad Hierarchy



Global Navigation **Menu 1:**

Local navigation:

- **Home**
 - **Store**
 - **Library**
 - **Merch**
 - **Wishlist**
 - **Events**
 - **Streaming**
- **Community**
 - **Forums**
 - **Reviews**
 - **Mods**
 - **Profile**
- **Support**
 - **Contact**
 - **About**
- **Login**
 - **Profile**

Supplementary navigation:

- **Search bar**

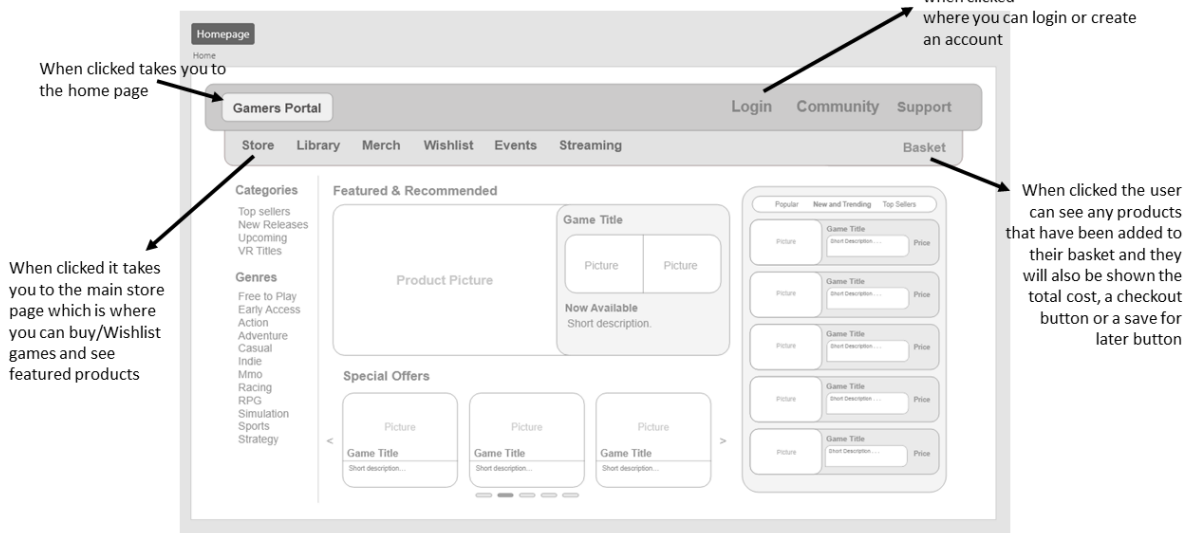
1.17 – 1.20 WIREFRAMING (MAXIMUM OF 5)

Complete this section in accordance with the coursework marking scheme.

Page Specification: No Page Specification.

Regular experience of Games Portal user

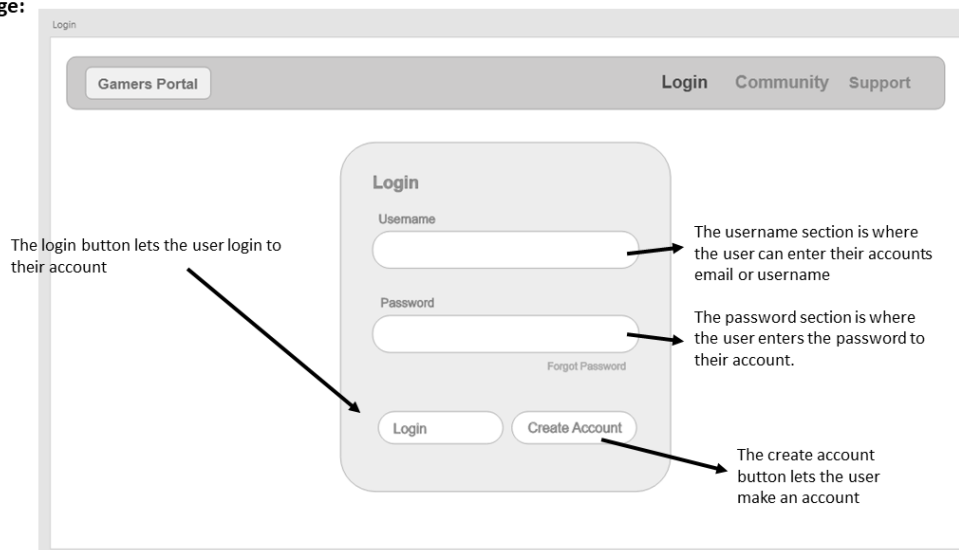
Home page:



This is the home page that is where the sites main content is displayed such as recommended games and offers as well as a second nav bar below the first that takes you to the site's main standout pages

Regular experience of Games Portal user

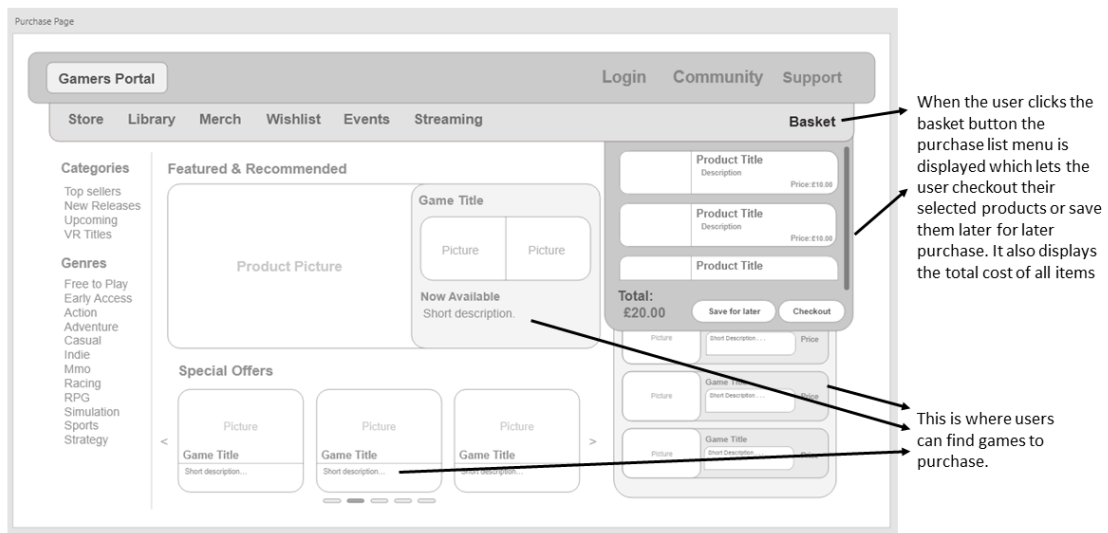
Login page:



This is the login page where the user can login to an existing account or create an account if they don't already have one.

Regular experience of Games Portal user

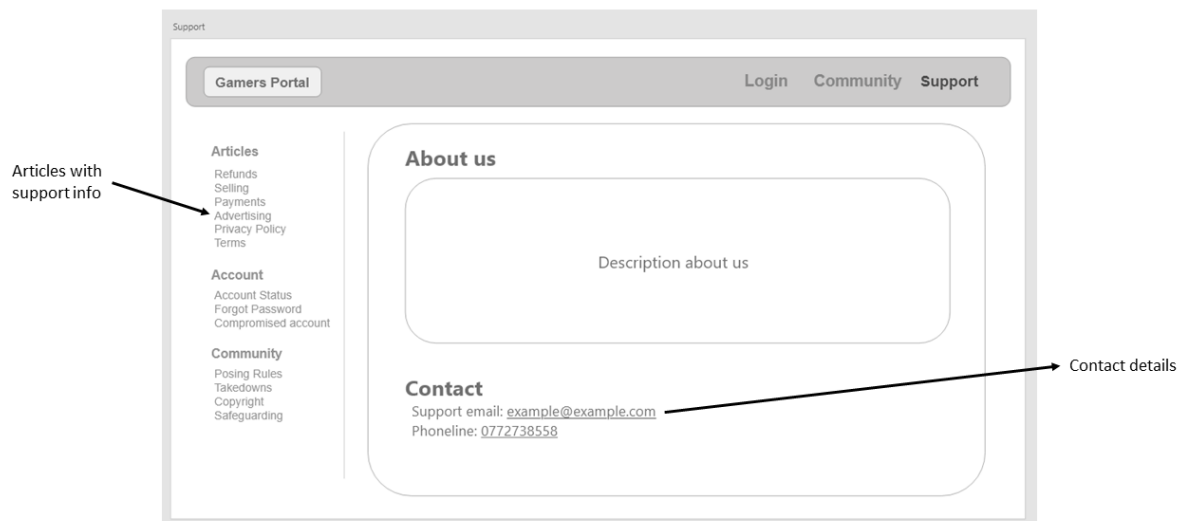
Purchase & Store page:



This is the purchase/store page where a menu is displayed to the user displaying any products they wish to buy or where users can browse through products that they wish to select for purchase.

Regular experience of Games Portal user

Support & about page:



If users have problems or concerns, they can navigate to this page where they can find support, contact details and articles related to any issues they face.